Temporal SCX: Audit Drift under Non-Stationary Distributions and the Sprint Self-Audit Limit

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Abstract

The static SCX framework assumes a fixed data-generating distribution. Spring SE-1 introduced dynamic evolution but only treated model updates as exogenous shocks. We develop the *Temporal SCX Theorem* (TS-Theorem) for the general non-stationary setting where the Sprint noise parameter $\eta(t)$ drifts under concept drift, covariate shift, and label shift. Three results are proved. (i) **Drift Detectability**: when the Situs distance between consecutive time slices exceeds a threshold $\delta_{\min}$, η-drift is detectable with power $\geq 1-\beta$ at significance α, with sample complexity $n_t + n_{t+1} \geq 2(\sigma_\xi^2 + \kappa\delta_{\min}^2)(z_{\alpha/2} + z_\beta)^2/\delta_{\min}^2$. This is **rigorous** — a direct consequence of T7 cross-domain transfer. (ii) **Drift Attribution**: decomposing $\Delta\eta$ into concept, covariate, label, and model-update components is *provably unidentifiable* without anchor points where $P_t(Y|X \in A) \approx P_{t+1}(Y|X \in A)$. With anchor points, $\Delta\eta_{\text{concept}}$ becomes estimable. This is an **open problem** in anchor construction. (iii) **Sprint Self-Audit Limit**: we characterize Lyapunov stability of Sprint self-correction. When $V(\eta) = (\eta - \eta^*)^2$ satisfies $\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] \leq \rho V(\eta_t)$ with $\rho < 1$ on a compact set, Sprint is geometrically ergodic and $\eta_t \to \eta^*$. However, when $v_{\text{drift}} > (1-\rho)/\Delta t$, the Lyapunov condition is violated and Sprint diverges — a **hard rate ceiling** on self-audit. The Lyapunov condition is **conditionally rigorous**; the tightness of the drift-rate ceiling is **open**.

1 Introduction

The SCX framework [?] was developed in a static setting: the Cercis Score $S(x) = Q(x) + \eta N(x)$ is computed once, and the audit is a snapshot. Spring SE-1 [?] introduced dynamic evolution but only treated model updates as exogenous shocks, leaving the data distribution stationary.

Reality is non-stationary. Concept drift ($P(Y|X)$ changes), covariate shift ($P(X)$ changes), and label shift ($P(Y)$ changes) are ubiquitous in deployed ML [? ?]. More fundamentally, the **Sprint parameter η itself drifts over time** — the coupling between noise and difficulty is not a constant but a function of the evolving data–model interaction.

This creates a recursive problem: **who audits the auditor?** Sprint (SE-1) calibrates η, but Sprint operates on data that is itself subject to drift. If η drifts silently, every noise flag, quality score, and gating decision rests on a miscalibrated foundation. TS-Theorem formalizes this self-referential audit problem.

Contributions.

(1) **Drift Detectability** (Theorem **??**): η-drift is detectable at controlled error rates given Situs distance between time slices. [Rigorous]

(2) **Drift Attribution** (Theorem **??**): Decomposition unidentifiable without anchors; estimable with them. [Open] (anchor construction)

(3) **Sprint Self-Audit Limit** (Theorem **??**): Sprint converges when $v_{\text{drift}} < (1-\rho)/\Delta t$, diverges otherwise. [Conditionally Rigorous] / [Open]

2 Preliminaries

2.1 Time-Indexed SCX

Definition 2.1 (Temporal SCX Process)**.** *At each discrete time $t = 0, 1, \ldots, T$:*

- *$P_t(X, Y)$ is the data-generating distribution;*
- *$S_t(x) = Q_t(x) + \eta_t N_t(x)$ is the Cercis Score;*
- *$U_t = \text{Situs}(P_t)$ is the Situs encoding;*
- *M_t is the deployed model;*

- *The Sprint parameter* $\eta_t \in [0, 1]$ *evolves as*

$$\eta_{t+1} = \eta_t + \Delta\eta(P_t, M_t) + \xi_t, \tag{1}$$

where $\Delta\eta(\cdot)$ *is the deterministic drift and* $\xi_t \sim (0, \sigma_\xi^2)$ *a random shock.*

The deterministic drift $\Delta\eta$ aggregates four sources:

$$\Delta\eta = \Delta\eta_{\text{concept}} + \Delta\eta_{\text{covariate}} + \Delta\eta_{\text{label}} + \Delta\eta_{\text{model}}, \tag{2}$$

corresponding to concept drift ($P(Y|X)$), covariate shift ($P(X)$), label shift ($P(Y)$), and model update ($M_t \to M_{t+1}$).

2.2 Situs Distance and Drift Coupling

From T7 (Cross-Domain Transfer Theorem), the Situs encoding is L_{Situs}-Lipschitz with respect to distributional change:

$$|\eta_{t+1} - \eta_t| \leq \kappa \cdot d_{\text{Situs}}(U_t, U_{t+1}) + |\xi_t|, \tag{3}$$

where $\kappa = L_{\text{Situs}} \cdot \sup_x N(x)$.

2.3 Anchor Points

Definition 2.2 (Concept-Stable Anchor Set). *A set* $A \subset \mathcal{X}$ *satisfies the* ***concept-stability condition*** *at times* $t, t+1$ *if*

$$\sup_{x \in A} \|P_t(Y \mid X = x) - P_{t+1}(Y \mid X = x)\|_{\text{TV}} \leq \varepsilon_A. \tag{4}$$

Anchors are instances whose true labels are known to be stable — e.g., curated reference samples or human-verified exemplars.

2.4 General Assumptions for Temporal SCX

Throughout the remainder of this paper, the following assumptions are adopted unless otherwise noted. Each theorem may invoke a subset and may add specialized conditions.

Assumption 2.3 (Compactness of Situs Space). *The Situs space* $(\mathcal{U}, d_{\text{Situs}})$ *is a compact metric space. Consequently, every sequence in* $\mathcal{U}$ *has a convergent subsequence, and the Wasserstein CLT of* **?** *] applies to empirical measures on* $\mathcal{U}$*.*

Assumption 2.4 (Regularity of Empirical Situs)**.** *Let $\widehat{U}_t$ be the empirical Situs encoding computed from n_t i.i.d. samples from P_t. There exists a function $\sigma_{\text{Situs}} : \mathcal{U} \times \mathcal{U} \to \mathbb{R}_+$ such that*

$$\sqrt{\frac{n_t n_{t+1}}{n_t + n_{t+1}}}\,\big(d_{\text{Situs}}(\widehat{U}_t, \widehat{U}_{t+1}) - d_{\text{Situs}}(U_t, U_{t+1})\big) \xrightarrow{d} \mathcal{N}\big(0,\ \sigma^2_{\text{Situs}}(U_t, U_{t+1})\big)$$

as $\min(n_t, n_{t+1}) \to \infty$, with the convention that $n_t/(n_t + n_{t+1}) \to \lambda \in (0, 1)$.

Assumption 2.5 (Lipschitz Coupling)**.** *Equation* (**??**) *holds with known constants $\kappa < \infty$ and $L_{\text{Situs}} < \infty$.*

Assumption 2.6 (Noise)**.** *The random shocks $\{\xi_t\}_{t\geq 0}$ are i.i.d. with $\mathbb{E}[\xi_t] = 0$, $\mathbb{V}[\xi_t] = \sigma^2_\xi < \infty$, and ξ_t independent of the data D_t.*

Assumption 2.7 (Anchor Availability)**.** *For Theorem* **??**, *there exists an anchor set $A \subset \mathcal{X}$ satisfying Definition* **??** *with $|A| = m$ and parameter $\varepsilon_A \ll 1$.*

Assumption 2.8 (Sprint Objective Smoothness)**.** *The expected audit error $\mathcal{E}(\eta)$ is twice differentiable on $[0, 1]$ with $\mathcal{E}''(\eta) \geq \mu > 0$ (strong convexity) and $|\mathcal{E}'(\eta)| \leq L_{\mathcal{E}}$ (Lipschitz gradient). There exists a unique minimizer $\eta^* \in (0, 1)$.*

3 Main Results

3.1 Drift Detectability

Theorem 3.1 (η-Drift Detectability)**.** *Let $d_t = d_{\text{Situs}}(U_t, U_{t+1})$. Assume Assumptions* **??**, **??**, **??**, *and* **??**. *If $d_t \geq \delta_{\min} > 0$, then at significance α, an auditor can detect η-drift with power $\geq 1 - \beta$ using combined sample size*

$$n_t + n_{t+1} \ \geq\ \frac{4(\sigma^2_\xi + \kappa^2 \cdot \delta^2_{\min})(z_{\alpha/2} + z_\beta)^2}{\delta^2_{\min}}, \tag{5}$$

where z_q are standard normal quantiles, σ^2_ξ is the random shock variance, and κ is the Situs-to-η Lipschitz constant from (**??**).

Proof. **.** SitusWald

Wald.

$$H_0 : \eta_t = \eta_{t+1} \quad \text{vs.} \quad H_1 : |\eta_{t+1} - \eta_t| \geq \kappa\, \delta_{\min}.$$

Situs

$$\widehat{d}_t = d_{\text{Situs}}(\widehat{U}_t, \widehat{U}_{t+1}),$$

$\widehat{U}_t n_t$i.i.d. $\{(X_i,Y_i)\}_{i=1}^{n_t} \sim P_t$Situs$\widehat{U}_{t+1} n_{t+1} \alpha H_0$

$$\widehat{d}_t > z_{\alpha/2}\,\frac{\sigma_d}{\sqrt{n_{\text{eff}}}},$$

$n_{\text{eff}} = n_t n_{t+1}/(n_t+n_{t+1})\sigma_d^2$

H_0. $H_0 : \eta_t = \eta_{t+1}$Situs$d_t = d_{\text{Situs}}(U_t, U_{t+1})\eta\eta\Delta_{\neg\eta}$Lipschitz (**??**)

$$0 = |\eta_{t+1} - \eta_t| \le \kappa\, d_t + |\xi_t| \quad\Longrightarrow\quad d_t \le \frac{|\xi_t|}{\kappa}.$$

$d_t = O_p(|\xi_t|) = O_p(\sigma_\xi)\sigma_\xi\delta_{\min} d_t H_0$

Assumption **??**Sommerfeld–MunkCLT

$$\sqrt{n_{\text{eff}}}\,\big(\widehat{d}_t - d_t\big) \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\text{Situs}}(U_t, U_{t+1})).$$

$H_0 d_t \to 0\mathbb{E}[\widehat{d}_t \mid H_0] \to 0$

$$\sqrt{n_{\text{eff}}}\,\widehat{d}_t \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\text{Situs}}(U_t, U_{t+1}) + \text{Var}(d_t)).$$

H_1. $H_1 : |\eta_{t+1} - \eta_t| \ge \kappa\,\delta_{\min}$Lipschitz

$$\kappa\,\delta_{\min} \le |\eta_{t+1} - \eta_t| \le \kappa\, d_t + |\xi_t| \quad\Longrightarrow\quad d_t \ge \delta_{\min} - \frac{|\xi_t|}{\kappa}.$$

$O_p(\sigma_\xi/\kappa) H_1 d_t \ge \delta_{\min}\widehat{d}_t$

$$\sqrt{n_{\text{eff}}}\,\big(\widehat{d}_t - d_t\big) \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\text{Situs}}(U_t, U_{t+1})).$$

$\mathbb{E}[\widehat{d}_t \mid H_1] \ge \delta_{\min} + o(1)$

σ_d^2. $\widehat{d}_t$

Iξ_tSitus. Situs$d_t = d_{\text{Situs}}(U_t, U_{t+1}) U_{t+1}\eta_{t+1} = \eta_t + \Delta\eta + \xi_t$Lipschitz

$$\frac{\partial d_t}{\partial \xi_t} = \frac{\partial d_t}{\partial U_{t+1}}\frac{\partial U_{t+1}}{\partial \eta_{t+1}}\frac{\partial \eta_{t+1}}{\partial \xi_t}.$$

SitusL_{Situs}-Lipschitz$\|\partial d_t/\partial U_{t+1}\| \le L_{\text{Situs}}$ Situs$\eta\gamma = \|\partial U/\partial\eta\|$ $\partial\eta_{t+1}/\partial\xi_t = 1$ $H_1 d_t \approx \delta_{\min}$Taylor

$$\text{Var}_\xi(d_t) \approx (L_{\text{Situs}}\gamma)^2\sigma_\xi^2.$$

$\kappa = L_{\text{Situs}} \cdot \sup_x N(x)\tilde{\gamma} = \gamma \cdot \sup_x N(x)$ $(L_{\text{Situs}}\gamma)^2 = \kappa^2(\gamma/\sup_x N(x))^2$ $\gamma \approx \sup_x N(x)$Situs$\eta N(x)\text{Var}_\xi(d_t) \approx \kappa^2\sigma_\xi^2$ $\sigma_\xi^2\kappa^2\delta_{\min}^2$Situs

IISitus. Sommerfeld–Munk CLTU_t, U_{t+1}

$$\mathrm{Var}(\widehat{d}_t \mid U_t, U_{t+1}) = \frac{\sigma^2_{\mathrm{Situs}}(U_t, U_{t+1})}{n_{\mathrm{eff}}}.$$

SitusAssumption **??** $\sigma^2_{\mathrm{Situs}}$Wasserstein [**?**] $d_t\sigma^2_{\mathrm{Situs}}(U_t, U_{t+1}) \propto d_t$ $H_1 d_t \approx \delta_{\min}$ $\sigma^2_{\mathrm{Situs}} \approx c \cdot \delta_{\min} c$ $c\kappa^2\sigma^2_{\mathrm{Situs}} \approx \kappa^2\delta^2_{\min}$

.

$$\sigma_d^2 = \sigma_\xi^2 + \kappa^2\delta^2_{\min}.$$

$\sigma_\xi^2\kappa^2\delta^2_{\min}$Situs $\sigma^2_{\mathrm{Situs}}\sigma_\xi^2 \ll \kappa^2\delta^2_{\min}$

.

$$\mathrm{Power}(\delta_{\min}) = \mathbb{P}\left(|\widehat{d}_t| > z_{\alpha/2}\, \frac{\sigma_d}{\sqrt{n_{\mathrm{eff}}}} \;\middle|\; H_1 \right).$$

$H_1\widehat{d}_t \approx d_t \geq \delta_{\min} n_{\mathrm{eff}}$

$$\mathrm{Power}(\delta_{\min}) \approx \Phi\left(\frac{\delta_{\min}\sqrt{n_{\mathrm{eff}}}}{\sigma_d} - z_{\alpha/2} \right),$$

Φ

$1-\beta$

$$\frac{\delta_{\min}\sqrt{n_{\mathrm{eff}}}}{\sigma_d} - z_{\alpha/2} \geq z_\beta.$$

$$n_{\mathrm{eff}} \geq \frac{(z_{\alpha/2} + z_\beta)^2\, \sigma_d^2}{\delta^2_{\min}}.$$

$n_{\mathrm{eff}} = n_t n_{t+1}/(n_t + n_{t+1})\sigma_d^2 = \sigma_\xi^2 + \kappa^2\delta^2_{\min}$ $n_t = n_{t+1} = n/2 n = n_t + n_{t+1}$ $n_{\mathrm{eff}} = n/4$

$$\frac{n}{4} \geq \frac{(z_{\alpha/2} + z_\beta)^2\, (\sigma_\xi^2 + \kappa^2\delta^2_{\min})}{\delta^2_{\min}},$$

$$n \geq \frac{4(\sigma_\xi^2 + \kappa^2\delta^2_{\min})(z_{\alpha/2} + z_\beta)^2}{\delta^2_{\min}}.$$

(**??**)□

□

Remark 3.2 (— TS.1)**.** *TS.1*

(1) Situs. *Sommerfeld–MunkWassersteinCLT [*__?__ *] $\mathcal{U}$Assumption* **??** *CLT SitusU*$_t$ = Situs(P_t)Situs $P_t\mathcal{U}$ *Wasserstein CLTβ-CLT*

(2) $\sigma_d^2 = \sigma_\xi^2 + \kappa^2 \delta_{\min}^2$. *$Cov(d_t, \widehat{d}_t - d_t)$*

$$\sigma_d^2 = \lim_{n\to\infty} n_{\mathit{eff}}\, \mathbb{V}\big(d_{\mathrm{Situs}}(\widehat{U}_t, \widehat{U}_{t+1})\big) = \lim_{n\to\infty} n_{\mathit{eff}} \left[\mathbb{V}(d_t) + \mathbb{E}[\mathbb{V}(\widehat{d}_t | U_t, U_{t+1})] + 2(d_t, \widehat{d}_t | U_t, U_{t+1})\right].$$

Sommerfeld–Munk CLT$\xi\eta \mapsto U$ JacobianSCX

(3) Lipschitz. $\kappa = L_{\mathrm{Situs}} \cdot \sup_x N(x)$ L_{Situs}*Situs Wasserstein-1MMD* $\sup_x N(x)N(x)$

(4) . $\Phi(-z_{\alpha/2} - \delta_{\min}\sqrt{n_{\mathit{eff}}}/\sigma_d)$ $\delta_{\min}\sqrt{n_{\mathit{eff}}}/\sigma_d \geq 310^{-3}$ $\delta_{\min} = \sigma_d n_{\mathit{eff}} = 4$ t
TS.1[Rigorous] SitusLipschitz

Applications: Theorem **??** provides a **principled alerting threshold** for production SCX monitoring. By tracking the empirical Situs distance $\widehat{d}_t$ between consecutive time slices, an operations team can set an alarm when d_t exceeds $\delta_{\min}$, triggering an audit of Sprint parameter drift. The sample-size formula (**??**) directly translates into monitoring SLAs: for a desired detection power of 0.95 at significance 0.05 with expected shift $\delta_{\min}$, the system must retain at least $n_t + n_{t+1}$ samples across the two time windows. In financial compliance systems where concept drift signals regulatory violations (e.g., money-laundering patterns evolving), this theorem provides a statistically-grounded early-warning system with guaranteed detection power. In recommendation systems, where user preference drift is continuous, the theorem can be used to schedule audit cycles: trigger an audit when the predicted Situs distance exceeds $\delta_{\min}$, rather than on a fixed calendar schedule.

Remark 3.3. *Theorem **??** is **rigorous**. It provides an operational criterion: when the Situs distance exceeds $\delta_{\min}$, the auditor has guaranteed power to detect the η shift. [Rigorous]*

3.2 Drift Attribution

Theorem 3.4 (Drift Source Attribution)**.** *Assume Assumptions **??** and **??**. The decomposition (**??**) into four source components is **unidentifiable** from Cercis Score observations alone: there exist distinct quadruples producing identical S_t, S_{t+1} distributions.*

*However, given an anchor set A satisfying Definition **??** with $|A| = m$, the concept drift component is independently estimable:*

$$\widehat{\Delta}\eta_{\mathrm{concept}} = \frac{1}{|\mathcal{X} \setminus A|} \sum_{x \notin A} \Delta S(x) - \frac{1}{|A|} \sum_{x \in A} \Delta S(x) + O\left(\varepsilon_A + \frac{1}{\sqrt{m}}\right). \tag{6}$$

The error bound holds with probability at least $1 - \delta$ for $m \geq 2\log(2/\delta)/\varepsilon_A^2$ (by Hoeffding inequality).

Proof. **Part (i): .**

. $tt+1$Cercis

$$\Delta S(x) = S_{t+1}(x) - S_t(x) = \Delta Q(x) + \eta_{t+1} N_{t+1}(x) - \eta_t N_t(x).$$

$\Delta\eta = \eta_{t+1} - \eta_t \Delta Q(x)$

$$\Delta Q(x) = \Delta Q_{\text{concept}}(x) + \Delta Q_{\text{covariate}}(x) + \Delta Q_{\text{label}}(x) + \Delta Q_{\text{model}}(x),$$

$P(Y|X)P(X)P(Y)$

$$\Delta\eta = \Delta\eta_{\text{concept}} + \Delta\eta_{\text{covariate}} + \Delta\eta_{\text{label}} + \Delta\eta_{\text{model}}.$$

. $\{\Delta S(x)\}_{x\in\mathcal{X}} |\mathcal{X}| \mathcal{X}$

$$\theta = (\Delta\eta_{\text{concept}}, \Delta\eta_{\text{covariate}}, \Delta\eta_{\text{label}}, \Delta\eta_{\text{model}})^\top \in \mathbb{R}^4.$$

$\theta \mathcal{L} : \mathbb{R}^4 \to \mathbb{R}^{|\mathcal{X}|} \Delta\eta \cdot \Delta N$

$$\Delta S(x) = \mathcal{L}(\theta)(x) = \sum_{j\in\{\text{concept},\dots\}} \frac{\partial S}{\partial \eta_j} \Delta\eta_j + \text{non-}\eta \text{ terms.}$$

$N(x) x \mathcal{L} 44$ -

$$\dim\ker(\mathcal{L}) = \dim \mathbb{R}^4 - \operatorname{rank}(\mathcal{L}) \geq 4 - 4 = 0.$$

ΔS $\operatorname{rank}(\mathcal{L}) \leq 1 \dim\ker(\mathcal{L}) \geq 3$

. $c \in \mathbb{R}$

$$\begin{aligned}
\Delta\eta'_{\text{concept}} &= \Delta\eta_{\text{concept}} + c, \\
\Delta\eta'_{\text{covariate}} &= \Delta\eta_{\text{covariate}} - \frac{c}{3}, \\
\Delta\eta'_{\text{label}} &= \Delta\eta_{\text{label}} - \frac{c}{3}, \\
\Delta\eta'_{\text{model}} &= \Delta\eta_{\text{model}} - \frac{c}{3}.
\end{aligned}$$

$|c|[0,1]$

$$\Delta\eta' = \Delta\eta + c - \frac{c}{3} - \frac{c}{3} - \frac{c}{3} = \Delta\eta.$$

$\Delta S(x) \Delta\eta \Delta Q \Delta\eta \Delta S \eta \Delta\eta$ $\Delta S'(x) = \Delta S(x) x$ 4Cercis

$N(x) x x$-ΔS3$\geq$ 1$\to$

Part (ii): .

. A (**??**)

$$\sup_{x\in A} \|P_t(Y|X=x) - P_{t+1}(Y|X=x)\|_{\mathrm{TV}} \le \varepsilon_A.$$

ACercis$O(\varepsilon_A)$

$$\Delta S_{\mathrm{concept}}(x) = O(\varepsilon_A), \quad \forall x \in A.$$

Cercis. $x \in A \Delta S(x) = \Delta S_{\neg\mathrm{concept}}(x) + O(\varepsilon_A)$ $\Delta S_{\neg\mathrm{concept}}$

$$\bar{\Delta}_A = \frac{1}{|A|}\sum_{x\in A}\Delta S(x) = \underbrace{\frac{1}{|A|}\sum_{x\in A}\Delta S_{\neg\mathrm{concept}}(x)} + O(\varepsilon_A).$$

$$\bar{\Delta}_{\mathrm{global}} = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}}\Delta S(x) = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}}\big(\Delta S_{\mathrm{concept}}(x) + \Delta S_{\neg\mathrm{concept}}(x)\big).$$

.

$$\widehat{\Delta\eta}_{\mathrm{concept}} = \frac{1}{|\mathcal{X}\setminus A|}\sum_{x\notin A}\Delta S(x) - \frac{1}{|A|}\sum_{x\in A}\Delta S(x).$$

. $\mu_{\mathrm{concept}} = \mathbb{E}[\Delta S_{\mathrm{concept}}(x)]$

$$\begin{aligned}\widehat{\Delta\eta}_{\mathrm{concept}} - \mu_{\mathrm{concept}} &= \big(\bar{\Delta}_{\mathcal{X}\setminus A} - \bar{\Delta}_A\big) - \mu_{\mathrm{concept}} \\ &= \underbrace{\big(\bar{\Delta}^{\mathrm{concept}}_{\mathcal{X}\setminus A} - \mu_{\mathrm{concept}}\big)} + \underbrace{\big(\bar{\Delta}^{\neg\mathrm{concept}}_{\mathcal{X}\setminus A} - \bar{\Delta}^{\neg\mathrm{concept}}_{A}\big)} + O(\varepsilon_A).\end{aligned}$$

1. $\bar{\Delta}^{\mathrm{concept}}_{\mathcal{X}\setminus A} - \mu_{\mathrm{concept}} = O_p(1/\sqrt{|\mathcal{X}\setminus A|}) \subset O(1/\sqrt{m}) |\mathcal{X}\setminus A| \gg m$
2. $\mathcal{X} O(1/m + 1/|\mathcal{X}\setminus A|) = O(1/\sqrt{m})$
3. $O(\varepsilon_A)$ **??**

Hoeffding$|\Delta S(x)| \le B \delta > 0$ $m \ge 2B^2 \log(2/\delta)/\varepsilon_A^2 \ge 1-\delta$

$$|\widehat{\Delta\eta}_{\mathrm{concept}} - \mu_{\mathrm{concept}}| \le C_1\varepsilon_A + C_2/\sqrt{m},$$

$C_1, C_2 m, \varepsilon_A$ (**??**)□

□

Remark 3.5 (— TS.2)**.** (1) . $c, -c/3, -c/3, -c/3$ $\Delta S(x)\Delta\eta$*Cercis* x-*Cercis* $\mathcal{L} : \mathbb{R}^4 \to \mathbb{R}^{|\mathcal{X}|}$ *1213* $|\mathcal{X}|$ $\{0\}$

$\Delta S(x)x$- *4*$k1 \leq k \leq 44 - k$ k*CercisTS.2*

(2) . $O(\varepsilon_A + 1/\sqrt{m})$ $\varepsilon_A \ll 1A$

- m
- ε_A
- $\bar{\Delta}^{\neg concept}_{\mathcal{X}\setminus A} \neq \bar{\Delta}^{\neg concept}_{A}$

¨ ¨*[Open]*

(3) . (**??**) -$P(Y|X)$ $P(X)$ (**??**) $O(\max_{i\neq j} \Delta\eta_i \cdot \Delta\eta_j)$

[Rigorous][Open]

Applications: Theorem **??** informs **drift triage** in production ML systems. When η-drift is detected (via Theorem **??**), the next question is "what changed?" The unidentifiability result means that, without anchor points, the operations team cannot determine whether the root cause is concept drift (requiring model retraining), covariate shift (requiring input monitoring adjustments), label shift (requiring recalibration), or model update effects (requiring rollback). The anchored estimator $\widehat{\Delta\eta}_{\text{concept}}$ provides a practical path forward: maintain a curated anchor set — a small collection of samples with verified stable labels — and use their Cercis Score changes to isolate the concept drift component. In autonomous driving perception systems, anchor points correspond to sensor readings under controlled conditions (e.g., clear weather, known road geometry) that serve as a stable reference for detecting environmental concept drift.

Remark 3.6. *The unidentifiability is a* ***rigorous*** *negative result. The anchor construction is an* ***open problem*** *— directionally clear but not yet formalized as an automated SCX procedure. [Open]*

3.3 Sprint Self-Audit Limit

Theorem 3.7 (Sprint Self-Audit Limit)**.** *Define $V(\eta_t) = (\eta_t - \eta^*)^2$ where η^* is the Sprint-optimal parameter. Let $\Omega \subset [0,1]$ be compact. Assume Assumption* **??** *and* **??**.

(i) Convergence. *If $\exists \rho < 1$ such that $\forall \eta_t \in \Omega$,*

$$\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] \leq \rho \cdot V(\eta_t), \tag{7}$$

then Sprint is ***geometrically ergodic****: $\mathbb{E}[(\eta_t - \eta^*)^2] \leq \rho^t(\eta_0 - \eta^*)^2$, and $\eta_t \to \eta^*$ in mean square.*

(ii) Divergence. *If the distribution drift rate* $v_{\text{drift}} = d_{\text{Situs}}(U_t, U_{t+1})/\Delta t$ *satisfies*

$$v_{\text{drift}} > \frac{1-\rho}{\Delta t}, \tag{8}$$

then the Lyapunov condition is violated and Sprint ***diverges****.*

Proof. **Part (i): .**

SprintLyapunov. Sprint

$$\eta_{t+1} = \eta_t - \lambda \nabla_\eta \mathcal{E}(\eta_t) + \xi_t,$$

$\mathcal{E}(\eta)\lambda > 0$ Lyapunov$V(\eta) = (\eta - \eta^*)^2$

$$\begin{aligned}
\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] &= \mathbb{E}\big[(\eta_t - \lambda\nabla\mathcal{E}(\eta_t) + \xi_t - \eta^*)^2 \mid \eta_t\big] \\
&= (\eta_t - \eta^*)^2 - 2\lambda(\eta_t - \eta^*)\nabla\mathcal{E}(\eta_t) + \lambda^2\|\nabla\mathcal{E}(\eta_t)\|^2 + \sigma_\xi^2 \\
&\quad + 2\mathbb{E}\big[(\eta_t - \lambda\nabla\mathcal{E}(\eta_t) - \eta^*)\xi_t \mid \eta_t\big] \\
&= (\eta_t - \eta^*)^2 - 2\lambda(\eta_t - \eta^*)\nabla\mathcal{E}(\eta_t) + \lambda^2\|\nabla\mathcal{E}(\eta_t)\|^2 + \sigma_\xi^2,
\end{aligned}$$

$\mathbb{E}[\xi_t \mid \eta_t] = 0\xi_t\eta_t$Assumption **??**

. Assumption **??**$\mathcal{E}'' \geq \mu > 0$

$$\nabla\mathcal{E}(\eta_t) \cdot (\eta_t - \eta^*) \geq \mu\,(\eta_t - \eta^*)^2, \qquad \|\nabla\mathcal{E}(\eta_t)\|^2 \leq L_\mathcal{E}^2\,(\eta_t - \eta^*)^2.$$

$$\begin{aligned}
\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] &\leq V(\eta_t) - 2\lambda\mu\,V(\eta_t) + \lambda^2 L_\mathcal{E}^2\,V(\eta_t) + \sigma_\xi^2 \\
&= \big(1 - 2\lambda\mu + \lambda^2 L_\mathcal{E}^2\big)\,V(\eta_t) + \sigma_\xi^2.
\end{aligned}$$

$\rho < 1$. $a(\lambda) = 1 - 2\lambda\mu + \lambda^2 L_\mathcal{E}^2$ $a(\lambda) = \rho < 1\sigma_\xi^2\Omega$
$a(\lambda)\lambda\lambda^* = \mu/L_\mathcal{E}^2$

$$a_{\min} = 1 - \frac{\mu^2}{L_\mathcal{E}^2}.$$

$\rho < 1\mu > 0\mathcal{E}\lambda = \lambda^*0 < \lambda < 2\mu/L_\mathcal{E}^2$ $\rho = a(\lambda^*) = 1 - \mu^2/L_\mathcal{E}^2 < 1$

Foster–Lyapunov. $\Omega\sigma_\xi^2$Lyapunov

$$\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] \leq \rho\,V(\eta_t) + \sigma_\xi^2$$

Meyn–TweedieFoster–Lyapunov [**?**]

Lyapunov$V\rho < 1C\eta \notin C$ $\mathbb{E}[V(\eta_{t+1})|\eta_t] \leq \rho V(\eta_t)V$-

$\sigma_\xi^2 \Omega 1$ Lyapunov

$$\mathbb{E}[V(\eta_t)] \le \rho^t V(\eta_0) + \frac{\sigma_\xi^2}{1-\rho}.$$

$t \to \infty \mathbb{E}[V(\eta_t)] \to \sigma_\xi^2/(1-\rho)$ $\sigma_\xi^2 = 0$Sprint$\mathbb{E}[(\eta_t - \eta^*)^2] \le \rho^t(\eta_0 - \eta^*)^2$ $\eta_t \to \eta^* \sqrt{\rho^t}$

Part (ii): .

. Sprintη^* $\eta_t^* \mathcal{E}_t(\eta) t$ $\eta_t^* P_t$

$$\Delta\eta_t^* = \eta_{t+1}^* - \eta_t^*$$

Lipschitz (**??**)Situs

$$|\Delta\eta_t^*| \le \kappa \cdot d_{\text{Situs}}(U_t, U_{t+1}) + O(|\xi_t|).$$

$O(|\xi_t|) v_{\text{drift}} = d_{\text{Situs}}(U_t, U_{t+1})/\Delta t$

$$|\eta_{t+1}^* - \eta_t^*| \approx \kappa \, v_{\text{drift}} \, \Delta t.$$

$v_{\text{drift}} \kappa v_{\text{drift}} := \kappa \cdot d_{\text{Situs}}(U_t, U_{t+1})/\Delta t$ $|\Delta\eta_t^*| \approx v_{\text{drift}} \, \Delta t$

Lyapunov. Lyapunov$t t+1 \eta_t^* \eta_{t+1}^* = \eta_t^* + \Delta\eta_t^*$

Sprint

$$\begin{aligned}\eta_{t+1} - \eta_{t+1}^* &= (\eta_t - \lambda \nabla \mathcal{E}_t(\eta_t) + \xi_t) - (\eta_t^* + \Delta\eta_t^*) \\ &= (\eta_t - \eta_t^*) - \lambda \nabla \mathcal{E}_t(\eta_t) + \xi_t - \Delta\eta_t^*.\end{aligned}$$

$\nabla \mathcal{E}$

$$\begin{aligned}\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] &= \mathbb{E}[(\eta_{t+1} - \eta_{t+1}^*)^2 \mid \eta_t] \\ &\approx \rho \, V(\eta_t) + (\Delta\eta_t^*)^2 - 2(\eta_t - \eta_t^*) \, \Delta\eta_t^*.\end{aligned}$$

$(\eta_t - \eta_t^*) \, \Delta\eta_t^* < 0$

$$\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] \ge \rho \, V(\eta_t) + (\Delta\eta_t^*)^2 - 2\sqrt{V(\eta_t)} \, |\Delta\eta_t^*|.$$

$|\Delta\eta_t^*| > (1-\rho)\sqrt{V(\eta_t)} V(\eta_t)$Lyapunov

$\Omega \sqrt{V(\eta_t)} \delta_{\min} > 0$ $|\Delta\eta_t^*| = v_{\text{drift}} \, \Delta t$

$$v_{\text{drift}} \, \Delta t > (1-\rho) \, \delta_{\min}.$$

$\delta_{\min} = 1 V \inf_{\Omega \setminus \{\eta^*\}} V \ge 1$

$$v_{\text{drift}} > \frac{1-\rho}{\Delta t}.$$

Lyapunov (**??**) Foster–Lyapunov $\mathbb{E}[V(\eta_t)]$Sprint

. Sprint Sprint$\rho(1-\rho)V(\eta_t)$ $v_{\text{drift}} \Delta t (v_{\text{drift}} \Delta t)^2$ Sprint□

□

Remark 3.8 (— TS.3)**.** *TS.3Markov*

(1) Lyapunov$V(\eta) = (\eta - \eta^*)^2$. *Foster–LyapunovV-aV*$\mathbb{E}[V(\eta_{t+1})|\eta_t] \leq \rho V(\eta_t) + b\mathbf{1}_{[]}C(\eta_t)$ *b*$C = \{\eta : V(\eta) \leq d\}$*dsmall set* $[0,1]$*Sprint* $\mathcal{E}\xi_t\psi$*- Foster–Lyapunov*$\xi_t\mathcal{E}$

(2) $v_{\mathrm{drift}} > (1-\rho)/\Delta t\kappa$. κv_{drift} *Lipschitz*κ κ

$$\kappa \cdot v_{drift} > \frac{1-\rho}{\Delta t}.$$

κ*Situs*η κ*Sprint SCX*

(3) $\nabla_\eta \mathcal{E}(\eta_t)$. $\mathcal{E}_t(\eta) = \mathbb{E}_{P_t}[L(\eta)]P_t$ $\widehat{\nabla \mathcal{E}}_t\delta_t = \widehat{\nabla \mathcal{E}}_t - \nabla \mathcal{E}_t$ *Lyapunov*

$$\mathbb{E}[V(\eta_{t+1})|\eta_t] = \rho V(\eta_t) + \lambda^2 \mathbb{E}[\|\delta_t\|^2|\eta_t] + \ldots.$$

$\lambda^2\|\delta_t\|^2$*Lyapunov*ρ*1*$\rho < 1$ *[Conditionally Rigorous]* $\sum_t \lambda_t\delta_t$

(4) . $v_{crit}\eta_t$ *[**?**]*$\mathcal{E}_t(\eta)$ *[Open]*

Lyapunov[Conditionally Rigorous] $\sigma_\xi^2\eta_t\eta^*$ κ*[Rigorous] [Open]*

Applications: Theorem **??** provides the **operational stability envelope** for Sprint self-calibration. The Lyapunov condition $\rho < 1$ translates to a concrete design requirement: the Sprint learning rate must dominate the Hessian of the expected audit error near η^*, ensuring that self-correction outpaces distribution drift. The divergence condition $v_{\mathrm{drift}} > (1-\rho)/\Delta t$ is the "red line": when the Situs distance between consecutive time slices exceeds this threshold, the operations team must either (i) reduce the audit cycle frequency (increase Δt), (ii) freeze η at its current value and switch to manual calibration, or (iii) deploy a higher-fidelity meta-auditor (reduce ρ via RA-Theorem's hierarchical audit architecture). In high-frequency trading systems, where v_{drift} can spike during market regime changes, this theorem provides the mathematical basis for circuit-breakers that temporarily suspend automated Sprint updates during periods of extreme non-stationarity.

Remark 3.9 (Status)**.** *The convergence is **conditionally rigorous**: given the Lyapunov condition, geometric ergodicity is standard Markov chain theory. The existence of Ω with $\rho < 1$ depends on Sprint's design. The divergence condition is a **rigorous** implication: if drift exceeds the threshold, stability fails. The* tightness *of the threshold is **open**. [Conditionally Rigorous] / [Open]*

Corollary 3.10 (Sprint Stability Window)**.** *Sprint self-audit operates stably only when $v_{\mathrm{drift}} \in [0, (1-\rho)/\Delta t)$. Monitoring v_{drift} via the Situs distance estimator provides an early-warning system: when v_{drift} approaches the boundary, intervention (slowing the audit cycle or freezing η) is required.*

Open Problem 3.11 (Tight Drift-Rate Threshold)**.** *Determine the* sharp *critical drift rate* v_{crit} *such that Sprint converges iff* $v_{\text{drift}} < v_{\text{crit}}$*. The Lyapunov bound gives* $v_{\text{crit}} \geq (1-\rho)/\Delta t$ *(a lower bound). The upper bound — and whether a sharp phase transition exists — is open, requiring analysis of Sprint as a stochastic approximation on a non-stationary objective [**?**].* *[Open]*

4 Comprehensive Strictness Critique

4.1

SCX

4.1.1 Sommerfeld–Munk Wasserstein CLT

SommerfeldMunk [**?**]WassersteinCLT $\mathcal{X}c(x,y)$ TS.1Situs$(\mathcal{U}, d_{\text{Situs}})$

- $\mathcal{U}$SCXSitus$\mathbb{R}^d$ $\mathcal{U}\mathbb{R}^d$Heine–BorelSitus Wasserstein
- **CLT** Wasserstein Donsker

SCXCercis$\mathcal{X}$ Situs$\mathcal{X}\mathcal{X}$

4.1.2

(**??**)

- $P(Y|X)P(X)$ $P(Y|X)P(X)$
- $P(Y|X) \propto P(Y)P(X|Y)$ $P(X|Y)$
- Sprint

TS.23$\mathcal{L}$21

4.1.3

TS.2

- $P_t(Y|X) \approx P_{t+1}(Y|X)$ $P(Y|X)$ Sprint
- $O(1/\sqrt{m})m$ $d \gg 1\mathcal{X}m$
- $P_t(X|A)A$

4.1.4 Lyapunov

TS.3Foster–Lyapunov

- $\{\eta : V(\eta) \leq d\}k$- Sprint$\eta_{t+1} = \eta_t - \lambda\nabla\mathcal{E}(\eta_t) + \xi_t$ ξ_t $\xi_t k > 1$
- $\Omega\partial\Omega$ Lyapunov$V\nabla\mathcal{E}$ Ω
- σ_ξ^2 ξ_t $\eta_t\eta^*\eta^*$ $\sigma_\xi^2/(1-\rho)$ “”“”

4.1.5 $\nabla_\eta\mathcal{E}(\eta_t)$

Sprint$\nabla_\eta\mathcal{E}(\eta_t)$

$$\widehat{\nabla\mathcal{E}}_t = \frac{1}{n_t}\sum_{i=1}^{n_t}\frac{\partial}{\partial\eta}\ell(\eta; x_i, y_i),$$

$\ell\delta_t = \widehat{\nabla\mathcal{E}}_t - \nabla\mathcal{E}_t$ $P_t n_t$ TS.3

- $\lambda^2\mathbb{E}[\|\delta_t\|^2]$Lyapunov
- $\delta_t\lambda(\eta_t - \eta^*)\mathbb{E}[\delta_t]$

aδ_tbδ_t c$\lambda\lambda_t \propto 1/\sqrt{t}1/t$ SCXλ

4.2

1. SitusSitus SitusWasserstein
2.
3. $\nabla\mathcal{E}$ Sprint
4. v_{crit} Benveniste [**?**] $\mathcal{E}_t$

5 Discussion

5.1 The Self-Referential Audit Structure

The three theorems reveal a self-referential structure:

(1) **Detection** (Theorem **??**): “Has something changed?” — answerable from Situs distances alone.

(2) **Attribution** (Theorem **??**): “What changed?” — requires anchor points external to the drift.

(3) **Limits** (Theorem **??**): "Can Sprint keep up?" — depends on drift rate vs. adaptation speed.

The self-referential tension is clearest in the limit theorem: Sprint uses η to calibrate the audit, η evolves with the data being audited, and the data evolves in response to the audit. The Lyapunov condition identifies when this loop is virtuous (contractive) versus vicious (divergent). When divergent, the sprint "outruns" its own calibration — a structural failure that no amount of data or computation can fix; only slowing the cycle or reducing v_{drift} restores stability.

5.2 Operational Implications

- **Monitor Situs distances.** Theorem **??** provides a principled alerting threshold.
- **Maintain anchor sets.** Anchor points are a theoretical necessity for diagnosability, not a luxury.
- **Track the drift rate.** When v_{drift} approaches $(1-\rho)/\Delta t$, intervention is required.

6 Conclusion

Temporal SCX extends static auditing to non-stationary distributions, revealing a self-referential structure where the auditor's calibration parameter η is embedded in the distribution being audited. Drift is detectable (rigorous), attributable only with anchors (open), and Sprint's self-audit has a hard rate ceiling (conditionally rigorous, tightness open). The resolution of Open Problem **??** — the sharp critical drift rate — would provide a design constraint for any deployed SCX system.

References

[1] SCX Working Group. "The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators," *Technical Report*, 2024.

[2] SCX Working Group. "Spring SE-1: Dynamic Evolution of the Sprint Parameter," *Technical Report*, 2024.

[3] J. Gama, I. Zliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia. “A survey on concept drift adaptation,” *ACM Computing Surveys*, 46(4):1–37, 2014.

[4] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang. “Learning under concept drift: A review,” *IEEE TKDE*, 31(12):2346–2363, 2018.

[5] S. P. Meyn and R. L. Tweedie. *Markov Chains and Stochastic Stability*, 2nd ed. Cambridge University Press, 2012.

[6] A. Benveniste, M. Métivier, and P. Priouret. *Adaptive Algorithms and Stochastic Approximations*. Springer, 1990.

[7] M. Zinkevich. “Online convex programming and generalized infinitesimal gradient ascent,” in *ICML*, 2003.

[8] M. Sommerfeld and A. Munk. “Inference for empirical Wasserstein distances on finite spaces,” *Journal of the Royal Statistical Society: Series B*, 80(1):219–238, 2018.

[9] G. I. Webb, R. Hyde, H. Cao, H. L. Nguyen, and F. Petitjean. “Characterizing concept drift,” *Data Mining and Knowledge Discovery*, 30(4):964–994, 2016.

[10] Z. C. Lipton, Y.-X. Wang, and A. Smola. “Detecting and correcting for label shift with black box predictors,” in *ICML*, 2018.

[11] T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[12] C. Villani. *Optimal Transport: Old and New*. Springer, 2008.